



INDIAN AIR FORCE

हर काम देश के नाम



INFORMATION HANDBOOK FOR SELECTION AS AGNIVEERVAYU

For more details

Log on to <https://agnipathvayu.cdac.in>



THE IAF MOTTO

'नमः स्पृशं दीप्तम्'

The motto of Indian Air Force, 'नमः स्पृशं दीप्तम्' has been taken from eleventh chapter of the Gita, the discourse given by Lord Krishna to Arjuna on the battlefield of Kurukshetra during the Great War of Mahabharata. The Lord is showing his Supreme Divine form to Arjuna and the great form of the Lord is reaching the sky with glory, evoking fear and loss of self-control in the mind of Arjuna. The Indian Air Force, similarly, aims to overwhelm the adversaries with application of aerospace power in defence of the nation.

**नमः स्पृशं दीप्तमनेकवर्णं व्यात्ताननं दीप्तविशालनेत्रम् ।
दृष्ट्वा हि त्वां प्रव्यथितान्तरात्मा धृतिं न विन्दामि शमं च विष्णो । ।**

Lord, seeing Your Form “Touching the Sky With Glory”, effulgent, multi-coloured, having its mouth wide open and possessing large flaming eyes, I, with my innermost self frightened, have lost self-control and find no peace.

Srimad Bhagvad Gita, Chapter XI, Verse 24

INDEX

Sl. No.	Subject	Page No.
1.	Brief History of the Indian Air Force	
2.	Eligibility Criteria	
3.	Medical Standards	
4.	Terms & Conditions	
5.	General Policy	
6.	Pay & Allowances and Perquisites	
7.	Types of Selection and Selection Procedure	
8.	Provisional Selection List (PSL)	
9.	Training	
10.	Career Progression	
11.	Address & contact details of Airmen Selection Centres	
12.	Syllabus and Model Question Paper of Physics	
13.	Syllabus and Model Question Paper of Mathematics	
14.	Syllabus and Model Question Paper of English	
15.	Syllabus and Model Question Paper of Reasoning and General Awareness	
16.	Syllabus and Model Question Paper for Education Instructor Trade	

BRIEF HISTORY OF THE INDIAN AIR FORCE

The Indian Air Force has had its share of trials and tribulation since its inception on 08 October 1932. The IAF's mettle was tested, to begin with, in the Second World War and thereafter, post-independence, in the conflicts of 1948, 1965, 1971 and in the Kargil conflict of 1999. The IAF has vindicated itself time and again by rising to the myriad challenges, tasks and demands placed on it by the Nation, within and outside India, whether during day or night. The fortitude and deeds of its men have won laurels both nationally and internationally. Its growth and continuous modernisation is an index of its ability to adapt quickly to changing times and technology. The deeds are testimony to the calibre of the air warriors.

IAF shed the prefix '**Royal**' when India became a Republic on 26 January 1950. To add to its pride, on 01 April 1954, one of its founder members, Air Marshal Subroto Mukherjee became the first Indian Chief of the Air Staff.

IAF added another feather in its cap when, the former Chief of the Air Staff, Air Chief Marshal Arjan Singh DFC was conferred with the rank of '**Marshal of the Indian Air Force**' on 26 January 2002.

ELIGIBILITY CRITERIA: AGNIVEERVAYU
(FOR BOTH MALE & FEMALE CANDIDATES)

AGE LIMIT AND EDUCATIONAL QUALIFICATION

	Age (As on date of Enrolment)	Educational Qualification
Science Subjects	17½ -21 Years	Candidates should have passed Intermediate/10+2/ Equivalent examination with Mathematics, Physics and English from an Education Board listed as COBSE member with minimum 50% marks in aggregate and 50% marks in English. OR Passed Three years Diploma Course in Engineering (Mechanical/ Electrical/ Electronics/ Automobile/ Computer Science/ Instrumentation Technology/ Information Technology) from a Government recognized Polytechnic institute with 50% marks in aggregate and 50% marks in English in diploma course (or in Intermediate/Matriculation, if English is not a subject in Diploma Course). OR Passed Two years Vocational course with non-vocational subject viz. Physics and Mathematics from State Education Boards/Councils which are listed in COBSE with 50% marks in aggregate and 50% marks in English in vocational course (or in Intermediate/ Matriculation, if English is not a subject in vocational course).
Other than Science Subjects	17½ -21 Years	Passed Intermediate / 10+2 / Equivalent Examination in any streams / subjects approved by Central / State Education Boards listed as COBSE member with minimum 50% marks in aggregate and 50% marks in English. OR Passed two years vocational course from Education Boards listed as COBSE member with minimum 50% marks in aggregate and 50% marks in English in vocational course or in Intermediate/Matriculation if English is not a subject in Vocational Course.

Note - 1: Candidates eligible for Science Subjects examination (Including Intermediate / 10+2 / three years' diploma course in engineering or two years' vocational course with non-vocational subjects of Physics and Maths) are also eligible for Other than Science Subjects and would be given an option of appearing in both Science Subjects and Other than Science Subjects examination in one sitting while filling up the online registration form.

Note - 2: Education Boards listed in Council of Boards for School Education (COBSE) website (cobse.org.in) as members, as on date of registration shall only be considered.

Note - 3: Exact aggregate Percentage of marks before decimal as written in the marks sheet of 10+2/Intermediate/Equivalent Examination/Three years Diploma Course/Two years Vocational Course OR calculated as per the rules of concerned Education Board / Polytechnic Institute shall only be considered (For example 49.99% should be taken as 49% and not to be rounded off to 50%).

MEDICAL STANDARDS

General Medical Standards for AGNIVEERVAYU are as follows:-

(a) **Height:**

- (i) **For Male candidates:-** Minimum acceptable height is 152.5 cms.
- (ii) **For Female candidates:-** Minimum acceptable height is 152 cms. For candidates belonging to North East or hilly regions of Uttarakhand, a lower minimum height of 147 cms will be accepted. In case, candidates from Lakshadweep, the minimum height will be 150 cms.

Note: Candidates availing height relaxation for being domicile of North East/Hilly Regions of Uttarakhand/Lakshadweep are to submit domicile certificate with endorsement of domicile status at the time of Phase-II testing. Further, candidates from hilly regions of Uttarakhand must have specific endorsement of hilly region in their domicile certificate.

(b) **Weight:** Proportionate to height and age.

(c) **Chest:** The chest wall should be well proportioned with minimum range of expansion of 05 cms.

(d) **Hearing:** Candidate should have normal hearing i.e. able to hear forced whisper from a distance of 6 meters with each ear separately.

(e) **Dental:** Should have healthy gums, good set of teeth and minimum 14 dental points.

(f) **Visual standards**

Visual Acuity	Maximum limits of Refractive error	Colour Vision
6/12 each eye, correctable to 6/6 each eye	Hypermetropia: +2.0D Myopia: 1D Including ± 0.50 D Astigmatism	CP-II

(g) **Corneal surgery (PRK/LASIK) is not acceptable.**

Note: Candidates should bring latest prescription and spectacle for corrected vision, if used. The prescription must bear diopter measurement, signature, stamp and registration number of Ophthalmologist. Date of prescription should be within a month prior to appearing for medicals.

(h) **General Health:** Candidate should be of normal anatomy without loss of any appendages. He / She should be free from any active or latent, acute or chronic, medical or surgical disability or communicable disease, infection and skin ailments. Candidate shall be physically and mentally FIT to perform duty in any part of the world, in any climate and terrain.

(j) **Gynaecological Examination (For female candidates):** The examination covers external genitalia, hernia orifices and the perineum, any evidence of stress urinary incontinence or genital prolapse outside introitus.

(k) **Gender (For female candidates):** Any candidate if found to have predominant characteristics of the opposite gender as evident on external physical examination, will be rejected as **Unfit**. Any candidate having undergone gender reassignment surgery will be declared **Unfit**.

(l) **Pregnancy (For female candidates):** Any candidate, if found to be pregnant shall be disqualified and her candidature rejected.

(m) **Body Tattoo:** Permanent body tattoos are not permitted, however, tattoos only on inner face of the fore arms (inside of elbow to the wrist), back (dorsal) part of the hand/reverse side of the palm and Tribals with tattoo which are as per the custom and traditions of their tribes may be considered. However, right to consider a tattoo acceptable or unacceptable will rest with the selection centre. **Candidates with permanent body tattoos are to submit two Postcard size photographs (close up and distinct view) with details of size and type of the Tattoo during the second phase of testing.**

(n) Details of medical standards are available on web portal <https://agnipathvayu.cdac.in>.

TERMS AND CONDITIONS

1. **AGNIVEERVAYU** will be enrolled in the Indian Air Force under Air Force Act 1950, for a period of four years. Only unmarried candidates are permitted to apply for enrolment in the IAF as Agniveervayu. Agniveervayu shall not be permitted to marry during their four years of service including training period. **AGNIVEERVAYU** would form a distinct rank in the Indian Air Force, different from any other existing ranks. Indian Air Force is not obliged to retain AGNIVEERVAYU beyond the engagement period of four years.
2. On completion of four years of service, based on organisation's requirements and policies promulgated by the Indian Air Force, AGNIVEERVAYU will be offered an opportunity to apply for permanent enrolment in the Indian Air Force. These applications will be considered in a centralised manner based on objective criteria including performance during their four year engagement period and up to 25% of each specific batch of AGNIVEERVAYU will be enrolled in regular cadre of IAF. **AGNIVEERVAYU will not have any right to be selected for further enrolment into the Indian Air Force. Selection of the AGNIVEERVAYU for further enrolment, if any, shall be at the discretion of the Government of India.**
3. **Employability.** AGNIVEERVAYU enrolled under this entry are liable to be assigned any duty in organisational interest, at the discretion of the Indian Air Force.
4. **Training.** On being enrolled, AGNIVEERVAYU will be imparted military training based on Indian Air Force requirement.
5. **Leave.** Grant of leave will be subject to exigencies of Indian Air Force. The following leave may be applicable for AGNIVEERVAYU during their engagement period:-
 - (a) **Annual Leave.** 30 days per year.
 - (b) **Sick Leave.** Based on medical advice.
6. **Medical and CSD Facilities.** For the duration of their engagement period in Indian Air Force, AGNIVEERVAYU will be entitled for medical facility at Service Hospitals as well as CSD provisions.

GENERAL POLICY

1. **Consumption of Narcotic Drugs and Psychotropic Substance (NDPS).** Consumption of Narcotic Drugs and Psychotropic substances banned under the NDPS Act 1985 **shall be a reject criteria for selection into the IAF** and AGNIVEERVAYU found either in possession or storing or distributing or consuming such drugs and substances after enrolment shall be liable for disciplinary action including dismissal from the IAF. **ZERO TOLERANCE POLICY TO ALCOHOLISM AND SUBSTANCE ABUSE IS FOLLOWED IN IAF.**
2. **Only Sikh candidates**, whose religion prohibits cutting of the hair or shaving of face, shall be permitted to grow hair and/or retain beard and moustache. Accordingly, those Sikh candidates willing to retain the same as per laid down specifications are to get their photographs with beard and moustache. Such candidates will not be permitted to grow / shave beard/moustache at later stage after enrolment.

PAY, ALLOWANCES AND ALLIED BENEFITS

1. AGNIVEERVAYU enrolled under Agnipath Scheme will be paid an Agniveer package of Rs. 30,000/- per month with a fixed yearly increment. In addition, Risk and Hardship allowances (as applicable in IAF), Dress and Travel allowances will be paid. **Perks such as ration, clothing, accommodation and Leave Travel Concession (LTC) shall also be provided as per the existing rules.**
2. **Terminal Benefits - Seva Nidhi Package.** AGNIVEERVAYU will be given one time 'Seva Nidhi' package comprising their monthly contribution along with matching contribution by the Government on completion of their engagement period, as indicated below:-

Year	Customised Package (Monthly)	In Hand (70%)	Contribution to Agniveers Corpus Fund (30%)	Contribution to Corpus fund by Gol
All Figures in Rs. (Monthly Contribution) (Approximately)				
1 st Year	30,000/-	21,000/-	9,000/-	9,000/-
2 nd Year	33,000/-	23,100/-	9,900/-	9,900/-
3 rd Year	36,500/-	25,550/-	10,950/-	10,950/-
4 th Year	40,000/-	28,000/-	12,000/-	12,000/-
All Figures in Rs. (Monthly Contribution) (Approximately)				
Total Contribution in Agniveers Corpus Fund after four years			Rs. 5.02 lakh	Rs. 5.02 lakh
Exit after 4 year	Approximately Rs. 10.04 Lakhs as Seva Nidhi Package (Absolute amount excluding interest)			

Note 1: AGNIVEERVAYU will not be required to contribute to any Provident Fund of the Government.

Note 2: There shall be no entitlement to gratuity and any kind of pensionary benefits in the case of AGNIVEERVAYU.

3. **Insurance, Death and Disability Compensation.** AGNIVEERVAYU will be provided non-contributory Life Insurance cover of Rs. 48 Lakhs for the duration of their engagement period as AGNIVEERVAYU in the Indian Air Force. AGNIVEERVAYU will not be governed by provisions contained in the Pension Regulations/Rules for Indian Air Force. For details of Death/Disability compensation log on to <https://agnipathvayu.cdac.in>.
4. **'Agniveer Skill Certificate'.** At the end of engagement period, a detailed Skill-Set certificate will be provided to the AGNIVEERVAYU, highlighting the skills and level of competency acquired by the personnel during their engagement period.
5. Personnel enrolled under this Scheme will not be eligible for Ex-Servicemen status.

TYPES OF SELECTION

The selection as an Agniveervayu in the IAF is conducted through Selection Test for Agniveervayu Recruitment (STAR) and Recruitment Rallies. The selection procedure of both types is given below:-

1. SELECTION PROCEDURE FOR SELECTION TEST FOR AGNIVEERVAYU RECRUITMENT (STAR):

Advertisements are published in Employment News and also uploaded on CASB Web portal <https://agnipathvayu.cdac.in> inviting online applications for STAR. In response to the advertisement, eligible male / female Indian Citizens are to apply online as per the instructions given in the web portal. The selection test will consist of following three stages:-

- (a) **Phase-I:** Eligible candidates will be sent provisional admit card for Phase-I of testing on their registered e-mail IDs. This provisional admit card is to be downloaded by the candidates under candidates login on CASB web portal <https://agnipathvayu.cdac.in>.

All candidates in possession of Provisional admit card will undertake online test as per their respective Subjects applied for, at centres designated/allotted as per their provisional admit card. Online test will be objective type and questions will be bilingual (English & Hindi) except for English paper. The result of Phase-I will be uploaded on <https://agnipathvayu.cdac.in>. The schemes of online examination is as given below:-

(i) **Science Subjects.** Candidates are tested in **English, Physics and Mathematics as per 10+2 CBSE syllabus.** Duration of written test will be **60 minutes.**

(ii) **Other than Science Subjects.** Candidates are tested in **English as per 10+2 CBSE Syllabus and Reasoning & General Awareness (RAGA).** Duration of written test will be **45 minutes.**

(iii) **Science Subjects and Other than Science Subjects.** Candidates are tested in **English, Physics & Mathematics of 10+2 CBSE syllabus and Reasoning & General Awareness (RAGA).** Duration of the written test will be **85 minutes.**

Marking Pattern for Written Test

- (i) 01 mark for every correct answer.
- (ii) Nil (0) mark for un-attempted question.
- (iii) 0.25 mark shall be deducted for each wrong answer or choosing more than one option as answer.

(b) **Phase-II:** Candidates shortlisted for Phase-II will be sent a new admit card on their registered e-mail IDs. Candidates have to report at the stipulated date and time for Phase-II at the designated ASC along with documents mentioned in admit card. The documents will be scrutinized/verified prior to commencement of Phase-II exam to ascertain the eligibility prima facie. Detailed verification of all the documents will be carried out in respect of candidates who pass Adaptability Test-1. Candidature of those who do not meet the laid down educational criteria is liable to be rejected at any stage of selection. The sequence of examination for Phase-II is as follows:-

(i) **Physical Fitness Test (PFT):**

(aa) **For Male candidates:-** 1.6 Km run to be completed within 07 minutes. Candidates will also have to complete 10 Push-ups, 10 Sit-ups and 20 Squats within the stipulated time to qualify in the Physical Fitness Test.

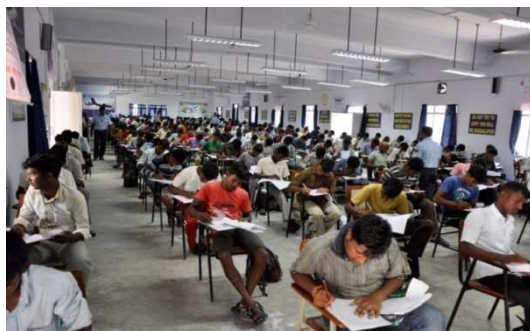
(ab) **For Female candidates:-** 1.6 Km run to be completed within 08 minutes. Candidates will also have to complete 10 Sit-ups and 15 Squats within the stipulated time to qualify in the Physical Fitness Test.

Note: Demonstrative videos of PFT uploaded on <https://agnipathvayu.cdac.in>



(ii) **Adaptability Test-1.** All the candidates who pass the Physical Fitness Test (PFT) will have to undertake Adaptability Test-1 (objective type written test) which is to assess suitability of a candidate for employment in the IAF which involves deployment in varied geographic terrain, weather and operational conditions.

(iii) **Adaptability Test-2.** All Candidates who pass Adaptability Test-1 will have to undertake Adaptability Test-2 as per policy in vogue. Adaptability Test-2 is to select candidates who can adapt to the environment of Indian Air Force and are able to adjust to the military way of life.



(c) **Phase-III (Medical Examination):** Candidates who qualify above tests will be medically examined at designated Medical Boarding Centre (MBC) on specified date. The Medical Examination will be conducted by Air Force Medical Team as per IAF medical standards and existing policy in vogue.

2. **SELECTION PROCEDURE FOR RECRUITMENT RALLIES**

Recruitment Rallies are conducted from time to time in selection areas/regions of particular States/Union Territories of the Country. Advertisement for Rallies are published with the details of eligibility condition, selection programme and rally venue in popular local/regional newspapers circulated in the region/area of the rally and also uploaded on website <https://agnipathvayu.cdac.in>. For Rally recruitment, eligible candidates are to report to the rally venue with requisite documents. These documents will be scrutinized and verified by the Testing Team. The sequence of examination is as follows:-

(a) **Physical Fitness Test (PFT):**

(i) **For Male candidates:-** 1.6 Km run to be completed within 07 minutes. Candidates will also have to complete 10 Push-ups, 10 Sit-ups and 20 Squats within the stipulated time to qualify in the Physical Fitness Test.

(ii) **For Female candidates:-** 1.6 Km run to be completed within 08 minutes. Candidates will also have to complete 10 Sit-ups and 15 Squats within the stipulated time to qualify in the Physical Fitness Test.



(b) **Written Test.** All candidates who clear the Physical Fitness Test (PFT) would appear in the written test as per the details given below:-

- (i) **Science Subjects.** Candidates are tested in **English, Physics and Mathematics as per 10+2 CBSE syllabus.** Duration of written test will be **60 minutes.**
- (ii) **Other than Science Subjects.** Candidates are tested in **English as per 10+2 CBSE Syllabus and Reasoning & General Awareness (RAGA).** Duration of written test will be **45 minutes.**
- (iii) **Science Subjects and Other than Science Subjects.** Candidates are tested in **English, Physics & Mathematics of 10+2 CBSE syllabus and Reasoning & General Awareness (RAGA).** Duration of the written test will be **85 minutes.**

Marking Pattern for Written Test

- (i) 01 mark for every correct answer.
- (ii) Nil (0) mark for un-attempted question.
- (iii) 0.25 mark shall be deducted for each wrong answer or choosing more than one option as answer.

(c) **Adaptability Test-1.** All candidates who pass the written test will have to undertake Adaptability Test-I (objective type written test) which is to assess suitability of a candidate for employment in the IAF which involves deployment in varied geographic terrain, weather and operational conditions.

(d) **Adaptability Test-2.** All Candidates who pass Adaptability Test-1 will have to undertake Adaptability Test-2 as per policy in vogue. Adaptability Test-2 is to select candidates who can adapt to the environment of Indian Air Force and are able to adjust to the military way of life.

(e) **Medical Examination :** Candidates who qualify above tests will be medically examined at designated Medical Boarding Centre (MBC) on specified date. The Medical Examination will be conducted by Air Force Medical Team as per IAF medical standards and existing policy in vogue.

Medical Examination would also include Baseline Investigation of:-

- (i) Blood Haemogram - Hb, TLC, DLC
- (ii) Urine RE/ME
- (iii) Biochemistry:-
 - (aa) Blood Sugar Fasting & PP
 - (ab) Serum Cholesterol
 - (ac) Urea, Uric acid, Creatinine
 - (ad) LFT-Serum Bilurubin, SGOT, SGPT
- (iv) X-Ray Chest (PA view)
- (v) Ultrasonography of lower abdomen & Pelvis (For female candidates only)
- (vi) ECG (R)
- (vii) Any other test necessary in the opinion of the medical officer.



PROVISIONAL SELECT LIST (PSL) FOR STAR AND RALLIES

Candidates who clear the online test/written test, PFT, Adaptability Test-1 and Adaptability Test-2 would figure in PSL. PSL will be displayed at all the Airmen Selection Centres (ASCs) and website <https://agnipathvayu.cdac.in>. Inclusion of names of candidates in the PSL depends upon the performance of the candidates in the exam. Inclusion of name in PSL does not guarantee automatic enrolment. Enrolment is strictly in order of merit subject to medical fitness, availability of vacancies, not exceeding the age of 21 years on date of enrolment and meeting all the laid down eligibility criteria as and when called for enrolment.

NOTE

1. The Recruitment test is not for selection of Officers/ Pilots/ Navigators.
2. For latest Eligibility Criteria (Date of Birth Block, Education Qualification and Medical Standards etc), Selection Procedure, Pay & Allowances and Perquisites log on to <https://agnipathvayu.cdac.in> and refer to IAF advertisements for Scheduled Tests/ Recruitment Rallies in Employment News and other National/ Regional Newspapers. For further assistance you may E-mail your queries to casbiaf@cdac.in.
3. Candidates should be prepared to stay for the test for four to five days under own arrangement. **No TA/DA shall be admissible.**
4. Selection in the Indian Air Force is absolutely free and fair. No money is required to be paid to any one for selection. Do not be misled by any one.

TRAINING

On enrolment, candidates shall be sent for a Joint Basic Phase Training (JBPT) at Basic Training Institute Belagavi (Karnataka). Training is however liable to be terminated at any time if the trainee fails to achieve the required standards in academics, profession, physical fitness and discipline or is found to be medically unfit **AFTER SUCCESSFUL COMPLETION OF ALL STAGES OF TRAINING, THE AGNIVEERVAYU SHALL BE DEPLOYED ON JOBS AT VARIOUS UNITS LOCATED PAN INDIA.**

CAREER PROGRESSION

On completion of four years of service, based on organisation's requirements and policies promulgated by the Indian Air Force, AGNIVEERVAYU will be offered an opportunity to apply for permanent enrolment in the Indian Air Force. These applications will be considered in a centralised manner based on objective criteria including performance during their four year engagement period and up to 25% of each specific batch of AGNIVEERVAYU will be enrolled in regular cadre of IAF.

ADDRESSES AND CONTACT DETAILS

1. Address, Contact numbers, website and E-Mail ID of CASB are as follows:-

CASB, AIR FORCE
AIR FORCE STATION NARAINA,
BRAR SQUARE, DELHI CANTT
NEW DELHI- 110 010
TELE – 011-25694209 /25699606
Website: <https://agnipathvayu.cdac.in>
E-Mail: casbiaf@cdac.in

2. There are **fourteen ASCs located all over the country.** Address, Contact numbers and E-Mail ID of ASCs are as follows:-

1 ASC, Air Force 48, Mansfield Road Ambala Cantt, Haryana - 133 001 Tele: 0171-2630048 e-mail: co.1asc-hr@gov.in	8 ASC, Air Force Air Force Station Tambaram Tambaram East, Chennai, Tamil Nadu- 600 046 Tele: 044-22390561 e-mail: co.8asc-tn@gov.in
2 ASC, Air Force Race Course Camp Air Force Station New Delhi New Delhi - 110 003 Tele: 011-23010231 Extn: 7751 e-mail: co.2asc-de@gov.in	9 ASC, Air Force Near Rajdhani College, Baramunda Bhubaneswar, Odisha -751003 Tele: 0674-2561336 e-mail: co.9asc-od@gov.in
3 ASC, Air Force Air Force Station Chakeri Kanpur, Uttar Pradesh - 208 008 Tele: 051-2405752 e-mail: co.3asc-up@gov.in	10 ASC, Air Force Air Force Station Bihta Patna, Bihar - 801 103 Tele: 06115-250001 Extn: 7762 e-mail: co.10asc-br@gov.in
4 ASC, Air Force Near Palta Gate Air Force Station Barrackpore 24-Parganas North, West Bengal-743 122 Tele: 033-25921251 Extn: 6391 e-mail: co.4asc-wb@gov.in	11 ASC, Air Force VIP Chowk, Borjhar Guwahati, Assam - 781 015 Tele: 0361-2842720 Extn: 333 e-mail: co.11asc-as@gov.in
5 ASC, Air Force Old Pali Road, Air Force Area Near Air Force Station Sub-Guard Room, Jodhpur, Rajasthan- 342 011 Tele: 0291-2511516 Extn: 7748, 7749 e-mail: co.5asc-rj@gov.in	12 ASC, Air Force Air Force Station Begumpet, Bowenpally, Secunderabad Telangana - 500 011 Tele: 040-27753512 e-mail: co.12asc-ap@gov.in
6 ASC, Air Force Air Force Station Cotton Green Mumbai, Maharashtra - 400 033 Tele: 022-23714982 Extn: 251 e-mail: co.6asc-mh@gov.in	14 ASC, Air Force VII/302-B, Vayu Sena Road Kakkanad, Kochi, Kerala - 682 030 Tele: 0484-2427010 e-mail: co.14asc-kr@gov.in
7 ASC, Air Force No. 1, Cubbon Road Bengaluru, Karnataka - 560 001 Tele: 080-25592199 e-mail: co.7asc-ka@gov.in	15 ASC, Air Force 2nd Floor, Phase-II, Rajiv Gandhi Parisar 35, Shyamala Hills, Bhopal Madhya Pradesh - 462 002 Tele: 0755-2661955 e-mail: co.15asc-mp@gov.in

SYLLABUS - PHYSICS

FOR SCIENCE SUBJECTS

1. Physical World and Measurement
2. Kinematics
3. Laws of Motion
4. Work, Energy and Power
5. Motion of System of Particles and Rigid Body
6. Gravitation
7. Properties of Bulk Matter
8. Thermodynamics
9. Behaviour of Perfect Gases and Kinetic Theory of Gases
10. Oscillations and Waves
11. Electrostatics
12. Current Electricity
13. Magnetic Effects of Current and Magnetism
14. Electromagnetic Induction and Alternating Current
15. Electromagnetic Waves
16. Optics
17. Dual Nature of Matter and Radiation
18. Atoms and Nuclei
19. Electronic Devices
20. Communication Systems

मॉडल प्रश्न पत्र
MODEL QUESTION PAPER

भौतिक विज्ञान
PHYSICS

- Q.1. एक पिण्ड त्वरण a से एक वृत्तीय पथ में गति कर रहा है। यदि उसका वेग दोगुना कर दिया जाए तो वेग बदलने के बाद तथा वेग बदलने से पहले, पिण्ड के त्वरणों का अनुपात ज्ञात करो।
A body is moving in a circular path with acceleration a . If its velocity gets doubled then find the ratio of acceleration after and before the change of velocity.
(A) 1:4 (B) 1:2 (C) 2:1 (D) 4:1

Ans : D

- Q.2. एक उपग्रह में घूमने वाले अंतरिक्ष यात्री की भारहीनता की स्थिति है -
Weightlessness of an astronaut moving in a satellite is a situation of -
(A) शून्य वेग / Zero velocity (B) कोई गुरुत्व नहीं / No gravity
(C) शून्य संहति / Zero mass (D) मुक्त रूप से गिरना / Free fall

Ans : D

- Q.3. ताप बढ़ाने पर, निम्नलिखित में से किस धातु का प्रतिरोध घटता है ?
For which of the following metals, the resistance decreases on increasing the temperature?
(A) ताँबा / Copper (B) टंगस्टन / Tungsten
(C) जर्मेनियम / Germanium (D) एल्युमीनियम / Aluminium

Ans : C

- Q.4. पृथ्वी के चुम्बकीय ध्रुवों पर नमन कोण क्या है ?
What is the angle of dip at magnetic poles of earth ?
(A) शून्य / Zero (B) 45° (C) 90° (D) 180°

Ans : C

- Q.5. एक आवेश चुम्बकीय क्षेत्र के लम्बवत् वृत्तीय पथ में गति करता है। इसके परिक्रमण का आवर्तकाल निम्न में से किस पर निर्भर नहीं करता है ?
A charge moves in a circle perpendicular to a magnetic field. Upon which of the following, the time period of revolution, does not depend?
(A) चुम्बकीय क्षेत्र / Magnetic field (B) आवेश / Charge
(C) कण का द्रव्यमान / Mass of the particle (D) कण का वेग / Velocity of the particle

Ans : D

- Q.6. एक नाभिक की परमाणु संख्या Z तथा परमाणु द्रव्यमान M है। न्यूट्रॉनों की संख्या ज्ञात करो।
Atomic number of a nucleus is Z and atomic mass is M . Find the number of neutrons.
(A) $M-Z$ (B) M (C) Z (D) $M+Z$

Ans : A

- Q.7. दिष्टकारी परिपथ में एक समान दिष्ट धारा प्राप्त करने हेतु उपयोग किये गये विद्युत परिपथ को ____ कहते हैं।
The electrical circuit, used to get smooth dc output from a rectifier circuit is called ____.
(A) दोलित्र / Oscillator (B) फिल्टर / Filter
(C) प्रवर्धक / Amplifier (D) लॉजिक गेट / Logic gate

Ans : B

- Q.8. 2 किग्रा व 4 किग्रा द्रव्यमान के दो पिण्ड क्रमशः 20 मी/से और 10 मी/से के वेग से परस्पर गुरुत्वाकर्षण के प्रभाव में एक दूसरे की ओर गतिशील हैं । इस निकाय के केन्द्रक द्रव्यमान का वेग, मी/से में ज्ञात करो ।
Two bodies of 2 Kg & 4 Kg are moving with velocities 20 m/s and 10 m/s respectively towards each other under mutual gravitational attraction. Find the velocity of their centre of mass in m/s.

(A) 5 (B) 6 (C) 8 (D) शून्य / Zero

Ans : D

- Q.9. 'r' त्रिज्या वाले ठोस गोले की एक निश्चित अक्ष के सापेक्ष घूर्णन त्रिज्या r है । इस अक्ष की गोले के केन्द्र से दूरी ज्ञात करो ।

The radius of gyration of a solid sphere of radius r about a certain axis is r . Find the distance of this axis from the centre of the sphere.

(A) r (B) $0.5r$ (C) $\sqrt{0.4} r$ (D) $\sqrt{0.2} r$

Ans : C

- Q.10. रुद्धोष्म प्रसार के सम्बन्ध में निम्न में से कौन सा कथन सही है ?

Which of the following statements is correct, in case of adiabatic expansion?

(A) $\Delta U = 0$ (B) $\Delta U =$ ऋणात्मक / negative (C) $\Delta U =$ धनात्मक / positive (D) $\Delta W = 0$

Ans : B

- Q.11. सरल आवर्त गति कर रहे किसी कण का वेग माध्य स्थिति पर _____ होता है ।

The velocity of a particle, executing S.H.M, is _____ at its mean position.

(A) अधिकतम / maximum (B) न्यूनतम / minimum
(C) अनन्त / infinity (D) शून्य / zero

Ans : A

- Q.12. 2 m^2 क्षेत्रफल की एक कुण्डली एक ऐसे चुम्बकीय क्षेत्र में रखी जाती है जो 2 सेकण्ड में 4 Wb/m^2 से परिवर्तित होकर 8 Wb/m^2 हो जाता है । कुण्डली में प्रेरित विद्युत वाहक बल ज्ञात करो ।

A coil of an area 2 m^2 is placed in a magnetic field which changes from 4 Wb/m^2 to 8 Wb/m^2 in 2 seconds. Find the induced e.m.f. in the coil.

(A) 4 V (B) 5 V (C) 6 V (D) 7 V

Ans : A

- Q.13. वह प्रक्रिया जिसके द्वारा प्रत्यावर्ती धारा को दिष्ट धारा में परिवर्तित किया जाता है, _____ कहलाती है ।

The process by which an alternating current is converted into direct current is called ____.

(A) शुद्धिकरण / Purification (B) आवर्धन / Amplification
(C) दिष्टकरण / Rectification (D) धारा आवर्धन / Current amplification

Ans : C

- Q.14. यदि सोडियम धातु पर प्रकाश वैद्युत प्रभाव के लिए देहली तरंगदैर्घ्य $5000 \text{ \AA}$ है तो इसका कार्यफलन ज्ञात करो ।

If the threshold wavelength for photoelectric effect on sodium metal is $5000 \text{ \AA}$ then find its work function.

(A) 15 J (B) $4 \times 10^{-19} \text{ J}$ (C) $4 \times 10^{-14} \text{ J}$ (D) $4 \times 10^{-22} \text{ J}$

Ans : B

- Q.15. रेडियो तरंगों को किस तरंग संचरण द्वारा एक स्थान से दूसरे स्थान तक भेजा जाता है ?
Through which mode of wave propagation, are the radio waves sent from one place to another?
(A) भू तरंग संचरण / ground wave propagation
(B) व्योम तरंग संचरण / sky wave propagation
(C) अंतरिक्ष तरंग संचरण / space wave propagation
(D) उपरोक्त सभी / all of the above

Ans : D

- Q.16. दृश्य प्रकाश की तरंगदैर्घ्य परास क्या है?
What is the wavelength range of visible light?
(A) $4 \times 10^{-7}m - 8 \times 10^{-7}m$ (B) $4 \times 10^{-6}m - 8 \times 10^{-8}m$
(C) $4 \times 10^5m - 8 \times 10^{-9}m$ (D) $4 \times 10^{10}m - 8 \times 10^{10}m$

Ans : A

- Q.17. सार्वत्रिक गुरुत्वाकर्षण नियतांक का विमीय सूत्र क्या है?
What is the dimensional formula for the universal gravitational constant?
(A) $M^{-1}L^3T^{-2}$ (B) $M^{-1}L^3T^{-1}$ (C) $M^{-1}L^2T^{-2}$ (D) $M^0L^0T^0$

Ans : A

- Q.18. दो गेंदों को क्रमशः h तथा 2h ऊँचाई से गिराया जाता है । उनके द्वारा जमीन तक पहुँचने में लिए गए समय का अनुपात क्या होगा ?
Two balls are dropped from heights h and 2h respectively. What would be the ratio of times taken by the balls to reach the earth ?
(A) $\sqrt{2} : 1$ (B) $1 : \sqrt{2}$ (C) 2:1 (D) 4:119

Ans : B

- Q.19. जब एक स्प्रिंग को 2 सेमी खींचा जाता है तो संचित ऊर्जा 100 जूल है । यदि उसे और 2 सेमी खींचा जाए तो ऊर्जा में कितने जूल की बढ़ोत्तरी होगी ?
When a spring is stretched by 2 cm, the energy stored is 100 J. If it is stretched further by 2 cm, its energy increases by _____.
(A) 300 (B) 400 (C) 200 (D) 100

Ans : A

- Q.20. पानी का पृष्ठ तनाव किस तापमान पर न्यूनतम होगा ?
At what temperature, will the surface tension of water, be minimum ?
(A) $0^\circ C$ (B) $25^\circ C$ (C) $60^\circ C$ (D) $75^\circ C$

Ans : D

- Q.21. जल की 2 बूंदों का व्यास क्रमशः 1 सेमी व 1.5 सेमी है । दोनों के अन्दर दाब आधिक्य का अनुपात ज्ञात करो
Diameters of 2 water drops are 1cm and 1.5 cm respectively. Find the ratio of excess pressures inside them.
(A) 1:1 (B) 5:3 (C) 3:2 (D) 2:3

Ans : C

- Q.22. यंग के द्विस्लिट प्रयोग में सोडियम प्रकाश ($\lambda = 5898 \text{ \AA}$) का प्रयोग करते हुए 92 फिजें प्राप्त होती हैं । यदि किसी अन्य प्रकाश ($\lambda = 5461 \text{ \AA}$) का प्रयोग करें तो प्राप्त फिजों की संख्या ज्ञात करो ।
In Young's double slit experiment, using sodium light ($\lambda = 5898 \text{ \AA}$), 92 fringes are seen. If another colour ($\lambda = 5461 \text{ \AA}$) is used then find the number of fringes.
(A) 62 (B) 99 (C) 67 (D) 85

Ans : B

- Q.23. दो समान्तर प्लेटों के विभव कमशः -10V एवं +30V हैं । यदि प्लेटों के बीच की दूरी 2 सेमी हो तो प्लेटों के मध्य विद्युत क्षेत्र ज्ञात करो ।
Two plates are at potentials -10V and +30V. If the separation between the plates is 2 cm then find the electric field between them.
(A) 2000 V/m (B) 1000 V/m (C) 500 V/m (D) 3000 V/m

Ans : A

- Q.24. किसी सूक्ष्मदर्शी में वस्तु को प्रदीप्त करने के लिए यदि लाल प्रकाश के स्थान पर नीले प्रकाश को प्रयुक्त किया जाए तो सूक्ष्मदर्शी की विभेदन क्षमता _____ ।
If red light is replaced by blue light illuminating the object in a microscope, the resolving power of the microscope _____.
(A) घट जाएगी / will decrease (B) बढ़, जाएगी / will increases
(C) आधी हो जाएगी / will get halved (D) अपवर्तित रहेगी / will remain unchanged

Ans : B

- Q.25. द्विपरमाण्विक गैस हेतु गैस के दो विशिष्ट ऊष्माओं का अनुपात $\frac{C_P}{C_V}$ का मान क्या होगा ?

In gases of diatomic molecules, Find the ratio of the two specific heat of gases $\frac{C_P}{C_V}$.
(A) 1.66 (B) 1.33 (C) 1.4 (D) 1.00

Ans : C

SYLLABUS - MATHEMATICS **FOR SCIENCE SUBJECTS**

1. Sets, Relations and Functions
2. Trigonometric Functions
3. Inverse Trigonometric Functions
4. Complex Number and Quadratic Equations
5. Permutations and Combinations
6. Binomial Theorem
7. Sequences and Series
8. Cartesian system of rectangular co-ordinates
9. Straight lines and family of lines
10. Circles and family of circles
11. Conic sections
12. Three-dimensional Geometry
13. Matrices and Determinants
14. Limit and Continuity
15. Differentiation
16. Applications of Derivatives
17. Indefinite integrals
18. Definite Integrals
19. Applications of the Integrals
20. Differential Equations
21. Vector
22. Probability
23. Statistics

मॉडल प्रश्न पत्र
MODEL QUESTION PAPER

गणित
MATHEMATICS

- Q.1. यदि सम्बन्ध R इस प्रकार परिभाषित है कि $R = \{(x, y) : 2x + y = 41, x, y \in \mathbb{N}\}$ तो R, निम्न में से किस प्रकार का सम्बन्ध है ?

What is the nature of relation R, if R is defined as $R = \{(x, y) : 2x + y = 41, x, y \in \mathbb{N}\}$?

- (A) स्वतुल्य / reflexive (B) सममित / symmetric
(C) संक्रामक / transitive (D) इनमें से कोई नहीं / None of these

Ans: D

- Q.2. $\cos 24^\circ + \cos 55^\circ + \cos 125^\circ + \cos 204^\circ + \cos 300^\circ = ?$

- (A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) 3 (D) 0

Ans: A

- Q.3. $\sec^{-1} \left[\frac{x^2 + 1}{x^2 - 1} \right] = ?$

- (A) $2 \tan^{-1} x$ (B) $2x^2$ (C) $2 \cot^{-1} x$ (D) x^2

Ans: C

- Q.4. अतिपरवलय $9x^2 - 16y^2 = 144$ की नाभियां ज्ञात करो ।

Find the foci of hyperbola $9x^2 - 16y^2 = 144$.

- (A) $(0, \pm 5)$ (B) $(\pm 5, 0)$ (C) $(\pm 5, 1)$ (D) $(5, \pm 1)$

Ans: B

- Q.5. उस त्रिभुज की प्रकृति ज्ञात करो जिसके शीर्ष बिन्दु $A(12, 8)$, $B(-2, 6)$ व $C(6, 0)$ हैं ।

Find the nature of the triangle whose vertices are $A(12, 8)$, $B(-2, 6)$ & $C(6, 0)$.

- (A) समद्विबाहु समकोणीय त्रिभुज / Isosceles Right angle triangle
(B) समबाहु त्रिभुज / Equilateral triangle
(C) विषमबाहु त्रिभुज / Scalene triangle
(D) इनमें से कोई नहीं / None of these

Ans: A

- Q.6. xy -प्लेन पर प्रत्येक बिन्दु $P(x, y, z)$ के लिए, ,
For every point $P(x, y, z)$ on the xy -plane,

- (A) $x = 0$ (B) $y = 0$ (C) $z = 0$ (D) None of these

Ans: C

- Q.7. $(6 + 5i)^2$ का संयुग्म ज्ञात करो ।
Find the conjugate of $(6 + 5i)^2$.

- (A) $60 + 11i$ (B) $11 - 60i$ (C) $11 + 60i$ (D) $60 - 11i$

Ans: B

- Q.8. $C(n, r) + 2C(n, r-1) + C(n, r-2) = ?$.

- (A) $C(n+1, r)$ (B) $C(n+2, r)$ (C) $C(n+2, r-1)$ (D) $C(n+1, r-1)$

Ans: B

- Q.9. एक गुणोत्तर श्रेणी का n वाँ पद 2^n है तो इसके प्रथम 6 पदों का योग ज्ञात कीजिए ।
 If n^{th} term of a G.P. is 2^n then find the sum of its first 6 terms.
 (A) 126 (B) 124 (C) 190 (D) 154

Ans: A

- Q.10. $\left[3x - \frac{1}{x}\right]^6$ के विस्तार में x^2 का गुणांक ज्ञात कीजिए ।
 Find the coefficient of x^2 in the expansion of $\left[3x - \frac{1}{x}\right]^6$
 (A) 405 (B) 7290 (C) 2430 (D) 1215

Ans: D

- Q.11. $\begin{bmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{bmatrix}^2 = ?$
 (A) $a^2b^2c^2$ (B) $4a^2b^2c^2$ (C) $\frac{1}{4}a^2b^2c^2$ (D) $(a+b+c)^2$

Ans: B

- Q.12. यदि $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ तो $A^{-1} = ?$

If $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, then $A^{-1} = ?$

- (A) A (B) $-A$ (C) I (D) $-I$

Ans: A

- Q.13. यदि ω ईकाई का घनमूल है तो $\begin{bmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{bmatrix} = ?$

If ω is the cube root of unity, then $\begin{bmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{bmatrix} = ?$

- (A) 1 (B) ω (C) ω^2 (D) 0

Ans: D

- Q.14. $\lim_{x \rightarrow 0} \frac{\sin(2+x) - \sin(2-x)}{x} = ?$

- (A) $\frac{1}{2}\cos 2$ (B) 1 (C) $2\cos 2$ (D) 0

Ans: C

Q.15. $\frac{d}{dx} \{\tan^{-1}(\sec x + \tan x)\} = ?$

- (A) $-\frac{1}{2}$ (B) 1 (C) -1 (D) $\frac{1}{2}$

Ans: D

Q.16. यदि $\sqrt{x+y} + \sqrt{y-x} = c$ तो $\frac{d^2y}{dx^2}$ ज्ञात करो ।

Find $\frac{d^2y}{dx^2}$, if $\sqrt{x+y} + \sqrt{y-x} = c$.

- (A) $\frac{2}{c}$ (B) $-\frac{2}{c^2}$ (C) $\frac{2}{c^2}$ (D) $\frac{4}{c^2}$

Ans: C

Q.17. एक घन की भुजा में 3 सेमी/सेकण्ड की दर से वृद्धि हो रही है । यदि घन की भुजा 10 सेमी है तो उसके आयतन में होने वाली वृद्धि की दर (सेमी³/सेकण्ड में) ज्ञात करो ।

An edge of a cube is increasing at the rate of 3 cm/sec. Find the rate at which does the volume increase (in cm³/sec) if the edge of the cube is 10 cm.

- (A) 900 (B) 725 (C) 700 (D) 825

Ans: A

Q.18. यदि $s = t^3 - 4t^2 + 5$ कण का गति बताता है और त्वरण लुप्त हो तो इसका वेग (ईकाई प्रति सेकण्ड) मे
If $s = t^3 - 4t^2 + 5$ describes the motion of a particle, then its velocity (in unit/sec) when the acceleration vanishes, is

- (A) $\frac{16}{9}$ (B) $-\frac{32}{3}$ (C) $\frac{4}{3}$ (D) $-\frac{16}{3}$

Ans: D

Q.19. संख्याओं 8, 12, 13, 15, 22 का मानक विचलन ज्ञात करो ।

Find the standard deviation of 8, 12, 13, 15, 22.

- (A) 3.54 (B) 3.72 (C) 4.21 (D) 4.6

Ans: D

Q.20. यदि एक सिक्के को तीन बार उछाला जाता है तो सिक्के में एक या दो शीर्ष आने की प्रायिकता ज्ञात करो ।

If a coin is tossed thrice, find the probability of getting one or two heads.

- (A) $\frac{4}{5}$ (B) $\frac{5}{8}$ (C) $\frac{3}{4}$ (D) $\frac{6}{7}$

Ans: C

Q.21. यदि बिन्दुओं $A(60\hat{i} + 3\hat{j})$, $B(40\hat{i} - 8\hat{j})$ और $C(a\hat{i} - 52\hat{j})$ समरेख हैं तो $a = ?$

If the points $A(60\hat{i} + 3\hat{j})$, $B(40\hat{i} - 8\hat{j})$ and $C(a\hat{i} - 52\hat{j})$ are collinear, then a is equal to

- (A) 40 (B) -40 (C) 20 (D) -20

Ans: B

Q.22. $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sin^2 x \, dx = ?$

- (A) 1 (B) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$ (C) $\frac{\pi}{2} - \frac{1}{4}$ (D) 0

Ans: B

Q.23. $\int \frac{\cos 2x}{\cos^2 x \sin^2 x} \, dx = ?$

- (A) $-\cot x - \tan x + c$ (B) $\cot x - \tan x + c$ (C) $\cot x + \tan x + c$ (D) $\tan x - \cot x + c$

Ans: A

Q.24. अवकल समीकरण $\frac{dy}{dx} = e^{x+y} + x^2 e^y$ का हल ज्ञात करो ।

Find the solution of the differential equation $\frac{dy}{dx} = e^{x+y} + x^2 e^y$.

- (A) $e^x + e^y - \frac{y^3}{3} = c$ (B) $e^x + e^y + \frac{x^3}{3} = c$ (C) $e^x + e^{-y} + \frac{x^3}{3} = c$ (D) $e^x + e^{-y} + \frac{y^3}{3} = c$

Ans: C

Q.25. वक्र $y^2 = 2y - x$ और y - अक्ष से परिवद्ध क्षेत्र का क्षेत्रफल (वर्ग मात्रक में) ज्ञात करो ।

Find the area of the region (in sq.units) bounded by the curve $y^2 = 2y - x$ & y - axis.

- (A) $\frac{8}{3}$ (B) $\frac{4}{3}$ (C) $\frac{5}{3}$ (D) $\frac{2}{3}$

Ans: B

SYLLABUS - ENGLISH

1. A short passage followed by four questions to test comprehension.

The questions will be set:

- (a) To judge understanding of the passage
- (b) To draw inference
- (c) To judge understanding of vocabulary

2. Grammar - 1

- (a) Subject - verb concord
- (b) Forms of verbs and errors in the use of verbs, etc.
- (c) Sequences of tenses and errors in the use of tenses
- (d) Transformation of sentences - Compound, Complex, Simple, Negative, Affirmative, Comparative degree, Positive degree, Superlative degree etc.

3. Grammar - 2

- (a) Formation of words - Nouns from verbs and adjectives, Adjectives from nouns and verbs, Adverbs from adjectives etc.
- (b) Determiners
- (c) The Preposition
- (d) Nouns and Pronouns
- (e) The Adjective
- (f) The Adverb
- (g) The Conjunction
- (h) The Modals
- (j) Clauses - Noun clauses, Adverb clauses of condition and time and Relative clauses

4. Vocabulary

- (a) Synonyms and Synonyms in context
- (b) Antonyms and Antonyms in context
- (c) One word substitution
- (d) Spelling pitfalls
- (e) Simple Idioms/phrases
- (f) Words often confused/Selecting the correct word fitting in a sentence.

5. **Narration (Direct and Indirect)**

- (a) Commands and requests
- (b) Statements (Various tenses)
- (c) Questions (Various forms of questions, tenses, etc.)

6. **Voice (Active and Passive)**

- (a) Changes required under each tense
- (b) Other conditions for transforming Active into Passive
 - (i) The Preposition
 - (ii) Modal auxiliaries
 - (iii) Infinitive
 - (iv) Participles

7. **Jumbled Sentences**

MODEL QUESTION PAPER

ENGLISH

Read the passage given below and answer the questions that follow: (Q.No.1 to 4):

Personally I dislike the praise of poverty and suffering. I do not think they are at all desirable, and they ought to be abolished. Nor do I appreciate the ascetic life as a social ideal though it may suit individuals. I understand and appreciate simplicity, equality, self-control but not the torturing of the body, I believe that the mind and habits have also to be trained and brought under control. It would be absurd to expect that a person who is given too much self-love can endure much suffering or show unusual self-control or behave like a hero when crisis comes. To be in good moral condition requires as much training to be in physical health.

Q.1. The writer does not like the praise of poverty. It is:

- (A) Personal opinion (B) Opinion of the all people
(C) Opinion of the ascetics (D) Opinion of the simple people

Ans : A

Q.2. To whom the ascetic life may be suitable?

- (A) To the writer (B) To the socially advanced
(C) To individuals (D) To all the members of society

Ans : C

Q.3. What does the author mean by 'ascetic life' in the passage?

- (A) Equality (B) Self-control
(C) Simplicity (D) Mortification of the body

Ans : D

Q.4. Which of the following words could replace the word 'absurd' as used in the passage?

- (A) reasonable (B) congruous (C) simple (D) silly

Ans : D

Identify the incorrectly spelt word:

Q.5. (A) Seperate (B) Occur (C) Receive (D) Accommodate

Ans : A

Choose the word almost nearest in meaning to the word given below:

Q.6. Zest

- (A) passion (B) apathy (C) lazy (D) boredom

Ans : A

Choose the word almost opposite in meaning to the word given below:

Q.7. Embellish

- (A) sacrifice (B) disfigure (C) forfeit (D) indict

Ans : B

Choose the correct one word substitute for the following stem:

Q.8. A well experienced person.

- (A) Officer (B) Senior (C) Apprentice (D) Veteran

Ans : D

A sentence, split into four parts, has been given. Choose the best order which produces the original sentence:

Q.9. for backward classes (1) / a vexed problem facing us (2) and to reserve more seats (3) / is the clamour to open more colleges (4)

- (A) 2,3,4,1 (B) 2,4,1,3 (C) 2,4,3,1 (D) 2,1,4,3

Ans : C

Choose the most appropriate answer and fill in the blanks: (Q.No.10 to 14)

- Q.10. The authorities seem to be _____ concerning the condition of workers.
(A) liberal (B) insensitive (C) ignorant (D) responsible **Ans : B**
- Q.11. He _____ for a walk daily in the morning.
(A) go (B) shall go (C) will go (D) goes **Ans: D**
- Q.12. _____ thing of beauty is a joy forever.
(A) A (B) The (C) One (D) An **Ans: A**
- Q.13. The thunder was accompanied _____ a heavy rain.
(A) from (B) up (C) by (D) with **Ans: C**
- Q.14. The noun form of 'detect' is _____.
(A) dictator (B) detention (C) detective (D) detection **Ans: D**

Choose the sentence that conveys the same meaning as the given sentence:

- Q.15. He is greater than me.
(A) I am not so great as he (B) I am the greatest
(C) I am as great as he (D) He is as great as I **Ans: A**

Spot the erroneous parts, if any, in the following sentences: (Q.No.16 to 18)

- Q.16. He met with an accident and admitted in the hospital. No error.
A B C D **Ans: C**
- Q.17. He stole a dog and sold the dog for Rs.50/- No error
A B C D **Ans: B**
- Q.18. Both Mili as well as Meena are beautiful. No error
A B C D **Ans: B**

Change the narration:

- Q.19. The teacher said to me, "Be regular and learn your lessons daily."
(A) The teacher told me to be regular and learn your lessons daily.
(B) The teacher asked me to be regular and learn my lessons daily.
(C) The teacher advised me be regular and learn my lessons daily.
(D) The teacher advised me to be regular and learn my lessons daily. **Ans: D**

Change the voice:

- Q.20. It is time to buy books.
It is time _____.
(A) all the books to be bought (B) for the books are being bought
(C) for the books to be bought (D) for the books have been bought. **Ans: C**

SYLLABUS - REASONING AND GENERAL AWARENESS

FOR OTHER THAN SCIENCE SUBJECTS

1. Reasoning (Verbal and Non-verbal)

- (a) Numerical Series
- (b) Distance and Direction Sense Test
- (c) Mathematical operations (Assigning Value to Arithmetic Sign)
- (d) Number, Ranking & Time Sequence Test
- (e) Sitting arrangements
- (f) Inserting the missing character
- (g) Human Relations
- (h) Coding and Decoding
- (j) Odd Man Out
- (k) Clocks
- (l) Tallest, Youngest Relations
- (m) Alphabet Test
- (n) Analogy
- (o) Non-Verbal Reasoning
- (p) Logical sequence of words
- (q) Logical Venn diagrams

2. Mathematics

- (a) Ratio and Proportion
- (b) Average
- (c) LCM, HCF
- (d) Profit and Loss
- (e) Time, Distance and Speed
- (f) Percentage
- (g) Simplification of Numbers
- (h) Fraction
- (j) Area of Triangle, Square, Circle and Rectangle
- (k) Surface area and Volume of Cuboids, Cylinder, Cone and Sphere
- (l) Probability
- (m) Simple Trigonometry

3. General Knowledge (GK) and Current Affairs (CA)

- (a) General Science
- (b) Civics
- (c) Geography
- (d) Current Events
- (e) History
- (f) Basic Computer Operations

मॉडल प्रश्न पत्र
MODEL QUESTION PAPER

तर्क शक्ति तथा सामान्य जागरूकता
REASONING AND GENERAL AWARENESS (RAGA)

- Q.1. 'रडार' द्वारा, किसी जहाज की स्थिति जानने हेतु, निम्न में से किन तरंगों का प्रयोग किया जाता है ?
Which type of waves are used by a 'RADAR' for detecting the presence of an aircraft?
(A) ध्वनि तरंगें / Sound Waves (B) रेडियो तरंगें / Radio Waves
(C) वैद्युत तरंगें / Electric waves (D) पराश्रव्य तरंगें / Ultrasonic Waves **Ans: B**
- Q.2. औरंगजेब द्वारा सिखों के कौन से गुरु को मृत्यु-दण्ड दिया गया था ?
Who among the following Sikh Gurus, was executed by Aurangzeb ?
(A) गुरु अर्जुन देव / Guru Arjun Dev (B) गुरु गोविन्द सिंह / Guru Govind Singh
(C) गुरु तेग बहादुर / Guru Tegh Bahadur (D) गुरु नानक देव / Guru Nanak Dev **Ans: C**
- Q.3. 'योगक्षेमं वहाम्यहम्', किस शासकीय नियंत्रित संस्था का आदर्श-वाक्य है ?
Which Government controlled organisation has its motto 'Yogakshemam Vahamyahum' ?
(A) भारतीय जीवन बीमा निगम / Life Insurance Corporation of India
(B) केन्द्रीय अन्वेषण ब्यूरो / Central Bureau of Investigation
(C) केन्द्रीय माध्यमिक शिक्षा बोर्ड / Central Board of Secondary Education
(D) भारतीय स्टेट बैंक समूह / State Bank of India Group **Ans: A**
- Q.4. _____ को 'हजार झीलों की भूमि' कहा जाता है।
_____ is called 'the land of thousand lakes'.
(A) स्विट्जरलैंड / Switzerland (B) स्वीडन / Sweden
(C) कनाडा / Canada (D) फिनलैंड / Finland **Ans: D**
- Q.5. मृत्यु दण्ड से क्षमादान का अधिकार, निम्न में से किसके पास है ?
Who among the following has been vested the exclusive power to grant pardon in case of death sentence.
(A) भारत के राष्ट्रपति / President of India
(B) भारत के प्रधानमंत्री / Prime Minister of India
(C) भारत के प्रधान न्यायाधीश / Chief Justice of India
(D) भारत के सॉलिसिटर जनरल / Solicitor General of India **Ans: A**
- Q.6. 'गर्बा' किस प्रदेश का प्रसिद्ध लोकनृत्य है ?
Of which of the following states, is 'Garba', a famous folk dance ?
(A) आन्ध्र प्रदेश / Andhra Pradesh (B) गुजरात / Gujarat
(C) बिहार / Bihar (D) पंजाब / Punjab **Ans: B**
- Q.7. अमेरिका के किस विशेष कार्य बल द्वारा, आतंकवादी ओसामा बिन-लादेन को खोजा व मारा गया ?
By which of the following Special Task Forces of USA, was the terrorist Osama Bin- Laden, searched & killed?
(A) मेरीन्स / Marines (B) हॉक / Hawks
(C) नेवी सील्स / Navy Seals (D) फॉक्स / Foxes **Ans: C**
- Q.8. यदि $\frac{5+2\sqrt{3}}{7+4\sqrt{3}} = a + b\sqrt{3}$ है तो a और b का मान ज्ञात कीजिए।
Find the value of a and b, if $\frac{5+2\sqrt{3}}{7+4\sqrt{3}} = a + b\sqrt{3}$.
(A) a = -11, b = -6 (B) a = -11, b = 6 (C) a = 11, b = -6 (D) a = 6, b = 11 **Ans: C**

- Q.9. दो स्टेशनों x और y के बीच की दूरी 778 किमी है । एक रेलगाड़ी x से y तक की दूरी 84 किमी/घण्टा की चाल से तथा वापसी की यात्रा 56 किमी/घण्टा की चाल से तय करती है । पूरी यात्रा के लिए, रेलगाड़ी की औसत चाल किमी/घण्टा में ज्ञात कीजिए ।
Distance between two stations x and y is 778 km. A train covers the journey from x to y at a speed of 84 kmph and returns to x with a speed of 56 kmph. Find the average speed of the train (in kmph) for the whole journey.
(A) 67.2 (B) 70 (C) 68 (D) 72 **Ans: A**
- Q.10. 50 प्रेक्षणों का माध्य 36 है । बाद में पता चला कि एक प्रेक्षण 48 को गलती से 23 ले लिया गया है । प्रेक्षणों का सही माध्य ज्ञात कीजिए ।
The average of 50 observations is 36. It was found later that an observation 48 was wrongly taken as 23. Find the correct average of the observations.
(A) 35.2 (B) 36.1 (C) 39.1 (D) 36.5 **Ans: D**
- Q.11. तीन संख्याओं का योग 136 है । यदि पहली और दूसरी संख्याओं का अनुपात 2:3 है और दूसरी और तीसरी संख्या का अनुपात 5 : 3 है तो दूसरी संख्या ज्ञात कीजिए ।
The sum of three numbers is 136. If the ratio between first and second is 2:3 and that between second and third is 5:3. Find the second number.
(A) 40 (B) 48 (C) 60 (D) 72 **Ans: C**
- Q.12. 45 व्यक्ति, एक काम को 16 दिन में कर सकते हैं । 6 दिन काम करने के बाद 30 व्यक्ति और शामिल हो जाते हैं । अब सभी को, बचा हुआ कार्य समाप्त करने में कितने दिन और लगेंगे ?
45 men can finish a task in 16 days. After 6 days, 30 more men join them. How many days, will they all, now take to complete the remaining task?
(A) 6 (B) 8 (C) 10 (D) 12 **Ans: A**
- Q.13. ज्ञात करो, किस राशि पर 10% वार्षिक ब्याज की दर से, 3 साल के चक्रवृद्धि ब्याज और साधारण ब्याज का अन्तर, ₹ 31 होगा ?
Find out the sum, for which the difference, between the compound and simple interest, for 3 years at the rate of 10% per annum, would be ₹ 31?
(A) Rs1500 (B) Rs 1200 (C) Rs 1600 (D) Rs 1000 **Ans: D**
- Q.14. वह न्यूनतम संख्या ज्ञात करो जिसे 8, 12, 20 या 25 से विभाजित करने पर प्रत्येक दशा में 5 शेष रहता हो ।
Find the smallest number, which when divided by 8, 12, 20 or 25 leaves a remainder 5 in each case?
(A) 595 (B) 605 (C) 1205 (D) 1195 **Ans: B**
- Q.15. एक आदमी अपनी आय का 75% खर्च करता है । यदि उसकी आय में 20% की वृद्धि, हो और खर्चों में 10% की वृद्धि, हो तो उसकी बचत में % वृद्धि ज्ञात कीजिए ।
A man spends 75% of his income. If his income increases by 20% and his expenditure increases by 10% then find the % increase in his savings.
(A) 40% (B) 30% (C) 50% (D) 60% **Ans: C**
- Q.16. विजातीय ज्ञात कीजिए ।
Find the odd one out.
(A) मिराज 2000 /Mirage 2000 (B) सुखोई-30 MKI/Sukhoi-30 MKI
(C) मिग-29/MiG-29 (D) मिग-27/MiG-27 **Ans: A**

Q.17. विजातीय ज्ञात कीजिए ।
Find the odd one out.

(A)



(B)



(C)



(D)



Ans:D

Q.18. यदि किसी कूट भाषा में '786' का अर्थ 'study very hard', '958' का अर्थ 'hard work pays' और '645' का अर्थ 'study and work' है तो उस कोड भाषा में 'very' का कूट ज्ञात करो ।
If, in a code language '786' means 'study very hard', '958' means 'hard work pays' and '645' means 'study and work' then find the code for 'very'?

(A) 8

(B) 7

(C) 6

(D) 9

Ans:C

Q.19. निम्नलिखित संख्या श्रृंखला को पूर्ण करें ।
Complete the following number series.
1, 4, 2, 8, 6, 24, 22, ____.

(A) 90

(B) 89

(C) 160

(D) 86

Ans:D

Q.20. एक परीक्षा, में एक छात्र ने प्रत्येक सही उत्तर के लिए 4 अंक प्राप्त किये व प्रत्येक गलत उत्तर के लिए 1 अंक खोया । यदि उसने सभी 75 प्रश्न हल किये और कुल 125 अंक प्राप्त किए तो उसके द्वारा, सही हल किये गए प्रश्नों की संख्या ज्ञात करो ।

In an examination, a student scores 4 marks for every correct answer and loses 1 mark for every wrong answer. If he attempts all 75 questions and secures 125 marks then find the number of questions, attempted correctly.

(A) 35

(B) 40

(C) 42

(D) 46

Ans:B

Q.21. दी गई सारणी में लुप्त संख्या ज्ञात करो ।
Find the missing number in the given table.

5	8	7
11	17	15
21	33	

(A) 29

(B) 31

(C) 28

(D) 33

Ans: A

Q.22. चार व्यक्ति A, B, C व D ताश खेल रहे हैं । A और B साझेदार हैं । यदि D का मुँह उत्तर की ओर हो तथा A का मुँह पश्चिम की ओर हो तो किसका मुँह दक्षिण की ओर होगा ?
Four persons A, B, C and D are playing cards. A & B are partners. If D is facing North and A is facing West, then whose face is towards South?

(A) C

(B) B

(C) D

(D) A

Ans: A

Q.23. निर्देश : उत्तर के रूप में सर्वोत्तम विकल्प का चयन करें ।

Direction : Choose the best alternative as the answer.

‘खतरे’ में हमेशा _____ होता है ।

Danger always involves_____.

(A) शत्रु /Enemy (B) हमला /Attack (C) डर /Fear (D) सहायता /Aid

Ans: C

Q.24. यदि + का अर्थ $\div$, $\div$ का अर्थ -, - का अर्थ $\times$ और $\times$ का अर्थ + है तो $8+2 \div 3 - 4 \times 6$ का मान ज्ञात करो ।

Evaluate $8+2 \div 3-4 \times 6$, if + means $\div$, $\div$ means $-$, $-$ means $\times$ and $\times$ means $+$.

(A) 0 (B) 1 (C) -2 (D) 3

Ans: C

Q.25. डॉक्टरों, वकीलों और पेशेवरों के समूह के बीच सही सम्बन्ध दर्शाने वाला वेन आरेख कौन सा है ?

Which of the following Venn diagrams depicts the correct relation among Doctors, Lawyers, Professionals?

(A)  (B)  (C)  (D) 

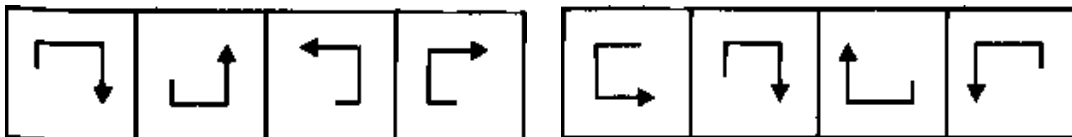
Ans: A

नीचे दिए गए प्रत्येक प्रश्न (प्रश्न संख्या 26-30) में, उत्तर चित्रों A, B, C तथा D में से वह चित्र चुनिये जो बाईं ओर दी गई श्रृंखला को पूर्ण करता है।

In each of the following Question (Q. No. 26-30), find the figure from the Answer figures A, B, C and D which would complete the series given on the left side.

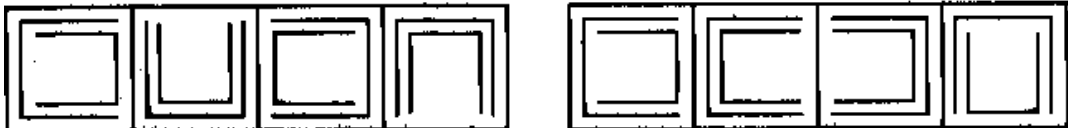
A B C D

Q.26.



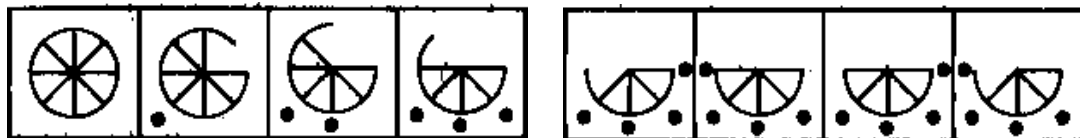
Ans: C

Q.27.



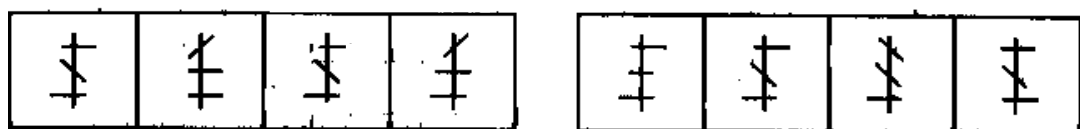
Ans: A

Q.28.



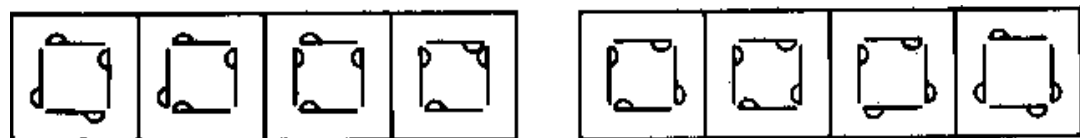
Ans: A

Q.29.



Ans: B

Q.30.



Ans: A

SYLLABUS - EDUCATION INSTRUCTOR
WRITTEN TEST OBJECTIVE (PART - I)

ENGLISH

I. A short passage followed by five questions to test comprehension.

The questions will be set:

1. To judge understanding of the passage.
2. To draw inference
3. To judge understanding of vocabulary

II. Grammar – 1

1. Subject – verb concord
2. Forms of verbs and errors in the use of verbs, etc.
3. Sequence of tenses and errors in the use of tenses
4. Transformation of sentences – Compound, Complex, Simple, Negative, Affirmative, Comparative degree, Positive degree, Superlative degree etc.

III. Grammar – 2

1. Formation of words - Nouns from verbs and adjectives, Adjectives from nouns and verbs, Adverbs from adjectives etc
2. Determiners
3. The Preposition
4. Nouns and Pronouns
5. The Adjective
6. The Adverb
7. The Conjunction
8. The Modals
9. Clauses - Noun clauses, Adverb clauses of condition and time and Relative clauses

IV. Vocabulary

1. Synonyms and Synonyms in context
2. Antonyms and Antonyms in context
3. One word substitution
4. Spelling pitfalls
5. Simple Idioms/phrases
6. Words often confused/Selecting the correct word fitting in a sentence

V. Narration (Direct and Indirect)

1. Commands and requests
2. Statements (Various tenses)
3. Questions (Various forms of questions, tenses, etc.)

VI. Voice (Active and Passive)

1. Changes required under each tense.
2. Other conditions for transforming Active into Passive
 - (a) The Preposition
 - (b) Modal auxiliaries
 - (c) Infinitive
 - (d) Participles

VII. Jumbled Sentences

1. To form a meaningful sentence
2. To form a meaningful paragraph

GENERAL KNOWLEDGE AND CURRENT AFFAIRS

1. General Reasoning
2. Elementary Maths
3. General Science
4. Civics
5. Geography
6. History
7. Basic Computer Knowledge
8. Current Affairs

SYLLABUS - EDUCATION INSTRUCTOR
WRITTEN TEST SUBJECTIVE (PART – II)

1. **Essay Writing.** An essay of about 500 words on any one topic out of four. The topics will be given on the following subjects.

- (a) Current Government policies
- (b) Current Socio-economic policies
- (c) Foreign policies
- (d) Science & Technology
- (e) Sports
- (f) Gender Related
- (g) Education related (policies, system etc)
- (h) Topics adapted from Daily Newspaper Editorials

2. **Precis Writing.** A passage of about 300 words will be given for precis writing with title.

MODEL QUESTION PAPER
WRITTEN TEST OBJECTIVE (PART - I)
ENGLISH

Read the passage give below and answer the questions that follow : (Q.No. 1 – 5)

If an opinion contrary to your own makes you angry, that is a sign that you are subconsciously aware of having no good reason for thinking as you do. If someone maintains that two and two are five or that Iceland is on the Equator, you feel pity rather than anger, unless you know as little of arithmetic or geography that his opinion shakes your own contrary conviction.

- Q.1. If someone else's opinion makes us angry, it means that
(A) we are subconsciously aware of having no good reason for becoming angry
(B) there may be good reasons for his opinion but we are not aware of them
(C) our own opinion is not based on good reason and we know this subconsciously
(D) we are not consciously aware of any reason for our own opinion.
- Q.2. "Your own contrary conviction" refers to
(A) the fact that you feel pity rather than anger
(B) the opinion that two and two are four and that Iceland is a long way from the Equator.
(C) the opinion that two and two are five and that Iceland is on the Equator
(D) the fact that you know so little about arithmetic or geography
- Q.3. Conviction means
(A) persuasion (B) disbelief (C) strong belief (D) ignorance
- Q.4. The writer says if someone maintains that two and two are five you feel pity because you
(A) have sympathy (B) don't agree with him
(C) want to help the person (D) feel sorry for his ignorance
- Q.5. The second sentence in the passage
(A) simply adds a further point to the argument already stated in the first sentence
(B) makes the main point which has only been introduced by the first sentence
(C) illustrates the point made in the first sentence
(D) builds up the argument of the first sentence by restating it from the opposite point of view

Choose the word that is furthest in meaning to the word given below:

- Q.6. Evasive
(A) Frank (B) Mistaken (C) Exuberant (D) Pensive

Choose the word that is most similar in meaning to the word given below:

- Q.7. Apposite
(A) Contrary (B) Bitter (C) Appropriate (D) Misleading

Choose the one which can be substituted for the given words/sentence:

- Q.8. Military waking signal sounded in the morning.
(A) Reveille (B) Lullaby (C) Anthem (D) Soprano

Choose the incorrectly spelt word:

- Q.9. (A) Hackneyed (B) Felicitious (C) Discourteous (D) Voluminous

Choose the correctly spelt word:

- Q.10. (A) Facsimile (B) Fascimilee (C) Fascimile (D) Facsimilee

Choose the correct meaning of the underlined phrase in the sentence below:

- Q.11. He is a silver tongued doctor and patients listen to him.
(A) speaking in a helpful but authoritative language
(B) speaking in a way that make people angry
(C) speaking in a way that bore people
(D) speaking in a way that persuade people

Choose the one which best expresses the given sentence in Passive voice:

- Q.12. Why did the teacher punish you?
(A) Why have you been punished by the teacher?
(B) Why had you been punished by the teacher?
(C) Why were you been punished by the teacher?
(D) Why were you punished by the teacher?

Choose the sentence which is most preferable in terms of grammar, meaning and usage:

- Q.13. (A) I wish I heard that story about him before I knew him.
(B) I wish I had heard that story about him before I knew him.
(C) I wish I had heard that story about him before I know him.
(D) I wish I heard that story about him before knowing him.

Read the dialogue given below and complete the passage in reported speech:

- Q.14. Interviewer : Was shooting in Kashmir challenging?
Santosh : Filming in any place is difficult but in Kashmir it was even more difficult because we had wanted to capture the first snowfall.
The interviewer asked Santosh _____ because they had wanted to capture the first snowfall.
(A) if shooting in Kashmir was challenging. Santosh replied that filming in any place was difficult but in Kashmir it had been even more difficult
(B) if shooting in Kashmir had been challenging. Santosh replied that filming in any place was difficult but in Kashmir it was even more difficult
(C) if shooting in Kashmir had been challenging. Santosh replied that filming in any place was difficult but in Kashmir it had been even more difficult
(D) if shooting in Kashmir was challenging. Santosh replied that filming in any place had been difficult but in Kashmir it was even more difficult

Detect the erroneous parts, if any, in the following sentences (Q.No. 15 – 17)

- Q.15. English spoken in India / cannot be fully / understood by an American. / No error.
A B C D

- Q.16. How long would it take / to swim / over the river? / No error.
A B C D

- Q.17. The First mammals have appeared on the earth approximately 100 million years ago.
A B C
No error.
D

In the following question, the first and the last words of a sentence/sentences of the passage are numbered 1 and 6. The rest of the sentence/passage is split into four parts named P,Q,R and S. Arrange these parts in proper order and choose the correct alternative from those given.

- Q.18. 1. It is far better to live for a short while
 P. contribution to the world
 Q. and make some significant
 R. that is just idled away
 S. than spend a long life
 6. in gossiping and playing.
 (A) RQSP (B) SQPR (C) QPSR (D) RQPS

In the following question, a part of the sentence is underlined. Choose the correct alternative to the underlined part which may improve the sentence. In case no improvement is needed your answer is (D).

- Q.19. I've met two women, whom, I believe, were hospital nurses.
 (A) whom I should have belief (B) who, I believe
 (C) whom, I am sure (D) No improvement

Given below is one sentence marked (A) after which there are five more sentences marked (B), (C), (D), (E) and (F). These sentences are not in their proper sequence. Find the proper sequence of the five sentences and mark your answer accordingly.

- Q.20. (A) The Delhi police had very little to do with kidnappings for ransom till about a decade ago.
 (B) All these traits are a boon for kidnappers.
 (C) An Anti-kidnapping cell has been created in the Crime Branch of the Delhi police.
 (D) Since then, however, Delhi has not only turned into a big money metropolis but also one which is extremely callous and indifferent.
 (E) But it has very little sophisticated gadgets or professionally trained men on hand to really make it effective.
 (F) The police on the other hand, has done little to hone its skills in dealing with this complex crime.
 The proper sequence should be
 (A) FCDBE (B) DBFCE (C) EBDCE (D) DBCEF

Fill in the blanks with the most appropriate choices : (Q.No.21 to 25)

- Q.21. On second reading, his poems strike us as singularly_____of sublime emotions.
 (A) Attributive (B) significative (C) symptomatic (D) evocative
- Q.22. In a classroom, students are to be trained to love _____
 (A) each other (B) all others (C) one another (D) altogether
- Q.23. I will phone you after _____
 (A) I shall have arrived (B) I arrive
 (C) I arrived (D) I shall arrive
- Q.24. We are looking forward_____you at next year's conference.
 (A) to see (B) seeing (C) that we will see (D) to seeing
- Q.25. It was_____project.
 (A) five-year (B) five-years (C) a five-years (D) a five-year

GENERAL KNOWLEDGE & CURRENT AFFAIRS

- Q.1. Complete the following number series.
1, 4, 2, 8, 6, 24, 22, 88, ____.
(A) 90 (B) 89 (C) 160 (D) 86
- Q.2. If, in a code language '786' means '**study very hard**', '958' means '**hard work pays**' and '645' means '**study and work**' then find the code for '**very**'?
(A) 8 (B) 7 (C) 6 (D) 9
- Q.3. 'Author' is related to 'Book' as 'Cobbler' is related to _____.
(A) Poem (B) Food (C) Meat (D) Shoes
- Q.4. A man spends 75% of his income. If his income increases by 20% and his expenditure increases by 10% then find the % increase in his savings.
(A) 40% (B) 30% (C) 50% (D) 60%
- Q.5. Distance between two stations **x** and **y** is 778 km. A train covers the journey from **x** to **y** at a speed of 84 kmph and returns to **x** with a speed of 56 kmph. Find the average speed of the train (in kmph) for the whole journey.
(A) 67.2 (B) 70 (C) 68 (D) 72
- Q.6. The average of 50 observations is 36. It was found later that an observation 48 was wrongly taken as 23. Find the correct average of the observations.
(A) 35.2 (B) 36.1 (C) 39.1 (D) 36.5
- Q.7. Which type of waves is used by a 'RADAR' for detecting the presence of an aircraft?
(A) Sound Waves (B) Radio Waves
(C) Electric waves (D) Ultrasonic Waves
- Q.8. The disease 'Beri-Beri' is caused by deficiency of _____.
(A) Vitamin K (B) Vitamin C
(C) Protozoa (D) Vitamin B
- Q.9. What is the chemical formula of common salt ?
(A) KCl (B) NaOH (C) NaCl (D) HCl
- Q.10. Who among the following has been vested the exclusive power to grant pardon in case of death sentence.
(A) President of India
(B) Prime Minister of India
(C) Chief Justice of India
(D) Solicitor General of India
- Q.11. Who administers the oath to the President of India?
(A) Lok Sabha Speaker (B) Vice President
(C) Chief justice of India (D) Prime Minister
- Q.12. _____ is called 'the land of thousand lakes'.
(A) Switzerland (B) Sweden
(C) Canada (D) Finland
- Q.13. Which one of the following rivers, does not originate in India ?
(A) Sutlej (B) Beas (C) Ravi (D) Chenab

- Q.14. Which state is the largest producer of pulses in India?
(A) Bihar (B) Rajasthan
(C) Madhya Pradesh (D) Maharashtra
- Q.15. Who among the following Sikh Gurus, was executed by Aurangzeb?
(A) Guru Arjun Dev (B) Guru Govind Singh
(C) Guru Tegh Bahadur (D) Guru Nanak Dev
- Q.16. Samrat Ashok was the ruler of which dynasty ?
(A) Maurya (B) Gupta (C) Pal (D) Kushan
- Q.17. The Red Fort was built by_____.
(A) Shahjaha (B) Akbar (C) Jahangir (D) Aurangzeb
- Q.18. What does the term 'E-mail' stands for?
(A) Electronic mail (B) Electric mail
(C) English mail (D) Emergency mail
- Q.19. By what name does the Indian supercomputer referred to as?
(A) SUPER 30 (B) PARAM
(C) MASTERMIND 50 (D) KALAM 5
- Q.20. What is a computer virus?
(A) A chemical that destroys computer
(B) A computer chip
(C) A computer programme
(D) A micro organism that destroys integrated circuits
- Q.21. Of which of the following states, is 'Garba', a famous folk dance ?
(A) Andhra Pradesh (B) Gujarat
(C) Bihar (D) Punjab
- Q.22. Who among the following has written the famous book 'Malgudi Days'?
(A) VS Naipaul (B) Deepak Chopra
(C) Rabindranath Tagore (D) RK Narayan
- Q.23. Who is the present chief minister of Tamil Nadu ?
(A) E Palaniswami (B) Banwarilal Purohit
(C) P Vijayan (D) None of these
- Q.24. Which country won the ICC 2019 cricket world cup?
(A) India (B) New Zealand
(C) Australia (D) England
- Q.25. Who has been appointed new chief of RAW ?
(A) Arvind Kumar (B) Krishnaswamy Natarajan
(C) Samant Kumar Goel (D) None of these

WRITTEN TEST – SUBJECTIVE (PART – II)

Q.1. Write an essay of about 500 words on any one of the following topics.

(Marks : 15)

- (a) Junk food is the Biggest Health Hazard for Young Generation.
- (b) Is India soft towards tackling terrorism? Comment.
- (c) Future of sports in India other than cricket.
- (d) Modernization of Indian Armed Forces.

Q.2. Write a précis of the following passage with a suitable title.

(Marks : 10)

Globalization is a phenomenon that has caught people flatfooted. Both the left and right in India attempt to catch it with old butterfly nets. For many in the left, globalization is 'late capitalism' and any text of Ernest Mandel or Prabhat Patnaik seems adequate to capture the canvas. The right flirts with the market and thinks it can control globalization through old mothballed concepts like swadeshi or a fundamentalism which is itself an ersatz phenomenon, producing a cheap soap opera when something more basic is required. We have handbooks on how to fight globalization before we sense what it is.

At another level, there is a metaphysical pathos about globalization, a sense that nothing can be done, that it is inevitable, that it can corrode a civilization in weeks. Such an approach dissolves the possibility of play and politics. If some writer strikes out huge parts of the map, or claims that Africa is doomed we listen seriously convinced that action and agency no longer belong to the African people. We triage them out of our virtual map.

There is then the question of metaphors. Is it a juggernaut, a seduction, a genocide, a set of flows, a master narrative? Each choice then produces its own repertoire of responses. Sadly we don't have the confidence of a Gandhi who when asked what he thought of western civilization said, 'That would be a good idea. 'The problem is we can't stop there for globalization is a bundle of ideas, a multiplicity of sites, a network of ambiguities. We have to pick up the good and bad ideas while this Brownean movement is in process. For now, globalization is best seen as a set of experimental sites, theatres, where the ambiguities, crosscurrents, countercurrents can be explored and spelt out. This issue is a small move in that direction.